GCSE BIOLOGY

Higher Tier – Paper 2 style

Topic 4.6: Inheritance, variation and evolution

Practice questions in the style of AQA GCSE Biology (8461). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 240 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **229**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 229

0 1 Organisms can reproduce sexually or asexually.

0 1 . 1 Complete the table to compare sexual reproduction and asexual reproduction. **[4 marks]**

	Sexual reproduction	**Asexual reproduction**
Number of parents		
Gametes involved?		
Genetic variation in offspring		

0 1 . 2 Name the type of cell division that produces gametes. **[1 mark]**

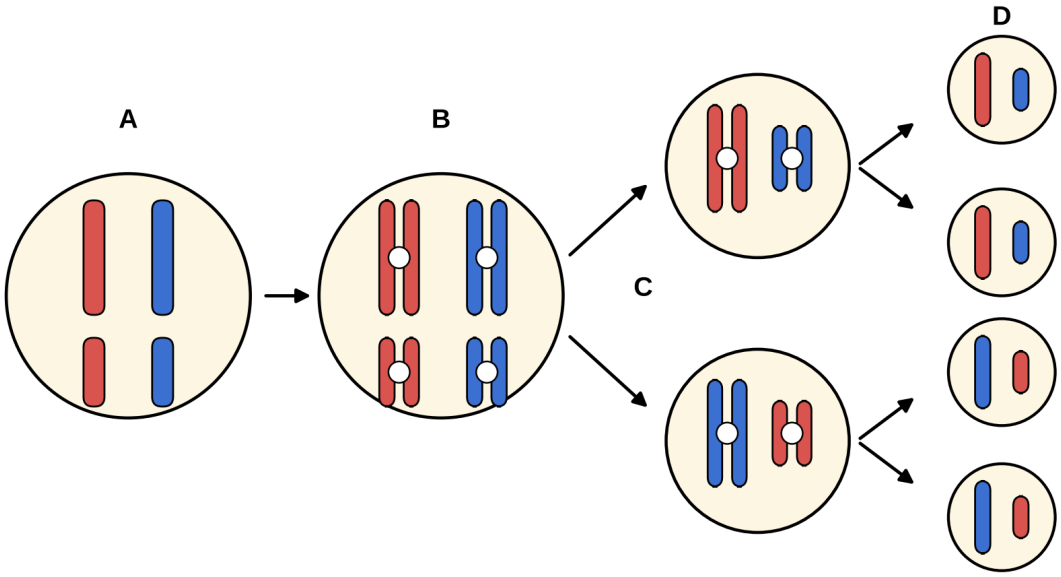
0 1 . 3 Give **two** advantages of sexual reproduction. **[2 marks]**

0 1 . 4 Give **two** advantages of asexual reproduction. **[2 marks]**

0 1 . 5 Some organisms can reproduce by both methods. Name **one** example and the way it can reproduce asexually. **[2 marks]**

0 1 . 6 Explain why fungi often reproduce asexually when conditions are good but sexually when conditions are poor. **[2 marks]**

0 2 The diagram shows an animal cell with four chromosomes dividing by meiosis. The letters **A** to **D** show the stages of the process.



0 2 . 1 What happens to the DNA between stage **A** and stage **B**? **[1 mark]**

0 2 . 2 How many chromosomes are in each cell at stage **D**? **[1 mark]**

0 2 . 3 A human body cell has 46 chromosomes. State how many chromosomes are in a human egg cell. **[1 mark]**

0 2 . 4 Describe what happens in meiosis to produce four cells that are genetically different from each other. **[3 marks]**

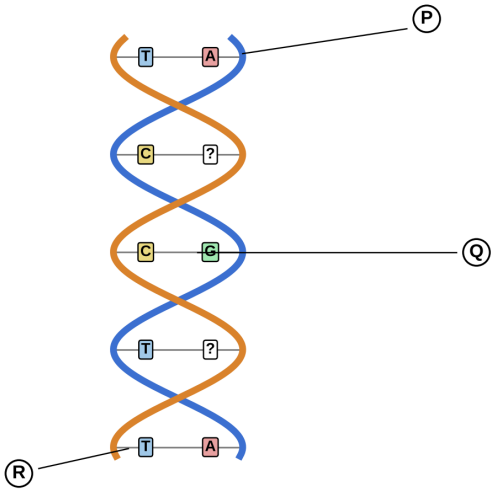
0 2 . 5 Gametes join at fertilisation. Describe what happens to the number of chromosomes at fertilisation and why this is important. **[2 marks]**

0 2 . 6 Explain how meiosis and fertilisation produce genetic variation. **[2 marks]**

0 2 . 7 Give **two** differences between mitosis and meiosis. **[2 marks]**

0 2 . 8 Name the parts of the body where meiosis happens in humans. **[1 mark]**

0 3 The diagram shows part of a DNA molecule. The letters **P**, **Q** and **R** show different parts. The base on one strand is shown on some of the rungs.



0 3 . 1 What is the shape of a DNA molecule? **[1 mark]**

0 3 . 2 Name the parts labelled **P**, **Q** and **R**. **[3 marks]**

0 3 . 3 State the base that pairs with **A** and the base that pairs with **C**. **[2 marks]**

0 3 . 4 DNA is a polymer made of repeating units. What are these units called? **[1 mark]**

0 3 . 5 What is the **genome** of an organism? **[1 mark]**

0 3 . 6 Explain why knowing the human genome is useful in medicine. **[3 marks]**

0 3 . 7 A gene contains 1200 bases. Each amino acid is coded by a sequence of three bases. Calculate the maximum number of amino acids in the protein. **[2 marks]**

Answer = _____

0 3 . 8 Describe how a gene codes for a protein. **[3 marks]**

0 3 . 9 A mutation changes one base in a gene. Explain why this may change the protein, but may not. **[3 marks]**

0 4 Genetics uses special terms.

0 4 . 1 Complete the table by writing the correct term for each description. Choose from the words in the box. **[6 marks]**

allele dominant genotype heterozygous homozygous phenotype recessive

Description	Term
a different version of the same gene	
an allele that is always expressed if present	
an allele that is only expressed if two copies are present	
the alleles a person has for a gene	
the characteristics that show in the organism	
having two identical alleles for a gene	

0 4 . 2 State the term used for an organism with two different alleles for a gene. **[1 mark]**

0 4 . 3 Name the part of a cell that contains the chromosomes. **[1 mark]**

0 4 . 4 A student says: 'A recessive allele can never affect the phenotype.' Explain why the student is wrong. **[2 marks]**

0 5 In humans the allele for brown eyes (**B**) is dominant to the allele for blue eyes (**b**). Two parents both have brown eyes and the genotype **Bb**.

0 5 . 1 Complete the Punnett square to show the possible genotypes of their children. **[2 marks]**

	B	**b**
B		
b		

0 5 . 2 What is the probability that a child will have blue eyes? **[1 mark]**

0 5 . 3 State the ratio of brown-eyed to blue-eyed children expected. **[1 mark]**

0 5 . 4 The parents have 80 children in total over many families. Calculate the number expected to have blue eyes. **[2 marks]**

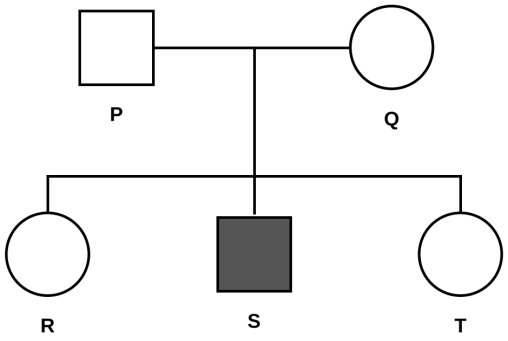
Answer = _____

0 5 . 5 A brown-eyed man and a blue-eyed woman have a blue-eyed child. State the genotype of the man and explain how you know. **[2 marks]**

0 5 . 6 A plant breeder has a tall pea plant. Tall (**T**) is dominant to dwarf (**t**). He crosses it with a dwarf plant and all the offspring are tall. What is the genotype of the tall plant? Give a reason. **[2 marks]**

0 5 . 7 A student says 'if two parents have brown eyes, all their children will have brown eyes.' Explain why this is not always true. **[2 marks]**

0 6 Cystic fibrosis (CF) is an inherited disorder caused by a recessive allele (**f**). The diagram shows a family. The shaded square shows a person with cystic fibrosis.



0 6 . 1 Give the genotype of person **S**. **[1 mark]**

0 6 . 2 Persons **P** and **Q** do not have cystic fibrosis. Explain how they can have a child who does. **[3 marks]**

0 6 . 3 Draw a Punnett square to show the possible genotypes of the children of **P** and **Q**. **[2 marks]**

0 6 . 4 Calculate the probability that the next child of **P** and **Q** will have cystic fibrosis. **[1 mark]**

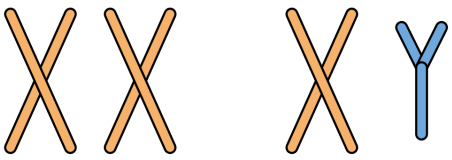
0 6 . 5 Person **R** does not have cystic fibrosis. Calculate the probability that **R** is a carrier. **[2 marks]**

0 6 . 6 About 1 in 25 people in the UK are carriers of the cystic fibrosis allele. Calculate the probability that a child of two unrelated people from the UK will have cystic fibrosis. **[3 marks]**

Answer = _____

0 6 . 7 Polydactyly (extra fingers or toes) is caused by a dominant allele. Explain why a person with polydactyly may have a child without the condition. **[2 marks]**

0 7 The diagram shows the sex chromosomes of two people, **J** and **K**.



Person J Person K

0 7 . 1 State which person is male and give a reason. **[2 marks]**

0 7 . 2 Draw a Punnett square to show how the sex of a child is determined. **[2 marks]**

0 7 . 3 State the probability that a child will be a girl. **[1 mark]**

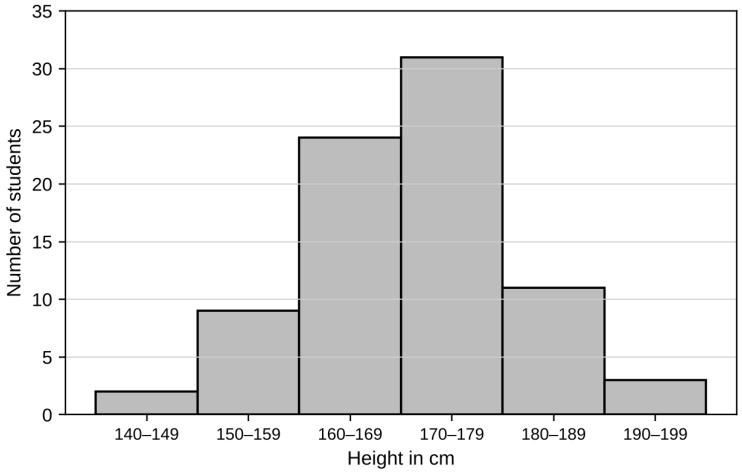
0 7 . 4 A couple have four girls. A friend says that their next child is more likely to be a boy. Is the friend correct? Explain your answer. **[2 marks]**

0 7 . 5 Calculate the probability that a couple will have three boys in a row. **[2 marks]**

Answer = _____

0 7 . 6 Explain why it is the father's sperm that determines the sex of the child. **[2 marks]**

0 8 A teacher measured the height of 80 students. The histogram shows the results.



0 8 . 1 State the modal class. **[1 mark]**

0 8 . 2 Calculate the percentage of students whose height is between 160 cm and 179 cm. **[3 marks]**

Answer = _____ %

0 8 . 3 Height shows continuous variation. What is meant by **continuous variation**? **[1 mark]**

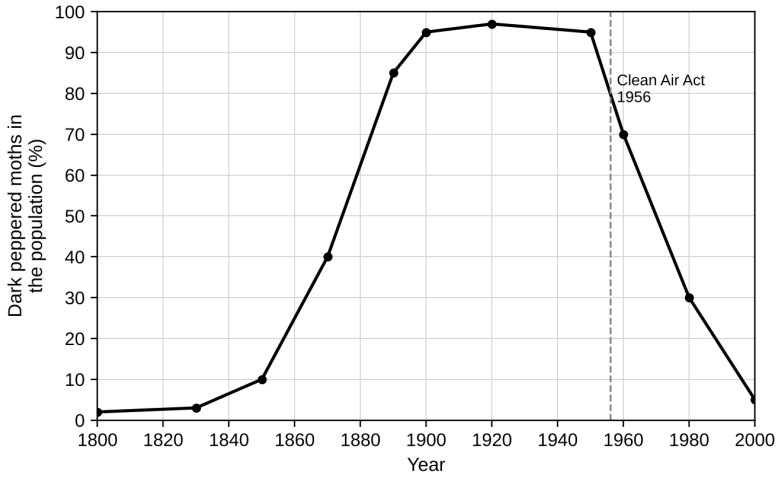
0 8 . 4 Give **two** causes of variation in height. **[2 marks]**

0 8 . 5 Identical twins are separated at birth and brought up in different places. One twin is 3 cm taller than the other. Explain why. **[2 marks]**

0 8 . 6 Give **one** characteristic caused only by genes and **one** caused by genes and environment. **[2 marks]**

0 8 . 7 Mutations are one source of variation. Explain how a mutation can lead to a new phenotype. **[3 marks]**

0 9 The peppered moth has pale and dark forms. Before 1850 most peppered moths were pale. The graph shows how the percentage of dark moths in a city changed. During the Industrial Revolution soot from factories blackened tree trunks. The Clean Air Act (1956) reduced the amount of soot in the air.



0 9 . 1 Describe the changes in the percentage of dark moths between 1850 and 1950. **[3 marks]**

0 9 . 2 Explain how natural selection caused the change in percentage of dark moths between 1850 and 1900. **[4 marks]**

0 9 . 3 Calculate the percentage decrease in the number of dark moths between 1960 and 2000. **[3 marks]**

Answer = _____ %

0 9 . 4 Suggest an explanation for the decrease in dark moths after 1956. **[2 marks]**

0 9 . 5 Explain how Charles Darwin's theory of evolution by natural selection is shown by this data. **[2 marks]**

0 9 . 6 Suggest **two** reasons why Darwin's theory was not accepted immediately. **[2 marks]**

1 0 This question is about evolution.

1 0 . 1 State what is meant by **evolution**. **[2 marks]**

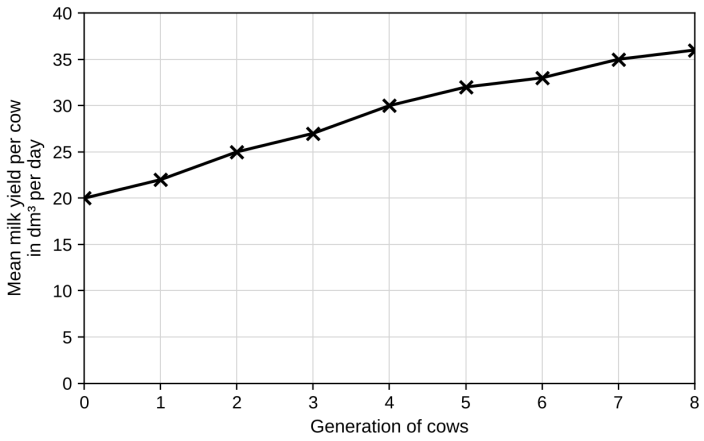
1 0 . 2 Describe how a new species may arise by natural selection. **[4 marks]**

1 0 . 3 Bacteria can evolve quickly. Explain how a population of bacteria becomes resistant to an antibiotic. **[3 marks]**

1 0 . 4 Explain why it is difficult to treat people infected with MRSA. **[1 mark]**

1 0 . 5 Give **two** ways of reducing the development of antibiotic-resistant strains. **[2 marks]**

1 1 Farmers use selective breeding to improve their cattle. The graph shows the mean milk yield of a herd of cows over eight generations.



1 1 . 1 What is meant by **selective breeding**? **[2 marks]**

1 1 . 2 Describe how the farmer would breed cows to increase milk yield. **[4 marks]**

1 1 . 3 Calculate the percentage increase in the mean milk yield from generation 0 to generation 8. **[3 marks]**

Answer = _____ %

1 1 . 4 Calculate the mean increase in milk yield per generation. **[2 marks]**

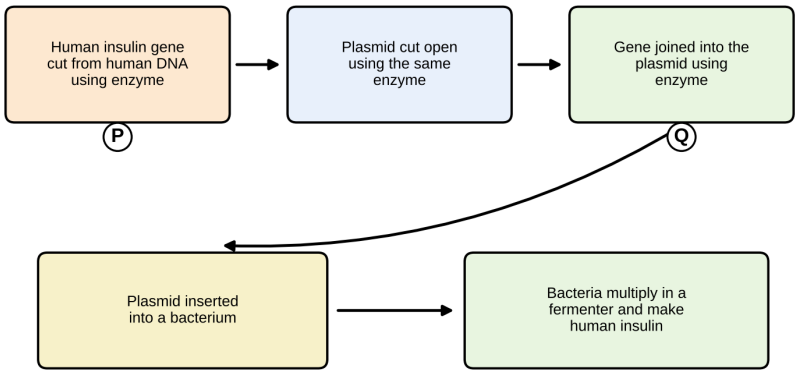
Answer = _____ dm^3/day

1 1 . 5 The increase in yield slows after generation 5. Suggest **one** reason. **[1 mark]**

1 1 . 6 Selective breeding can cause inbreeding. Explain the problems this can cause. **[2 marks]**

1 1 . 7 Give **two** examples of characteristics selectively bred in plants and animals, other than milk yield. **[2 marks]**

1 2 Genetic engineering is used to make human insulin. The diagram shows how it is done.



1 2 . 1 What is meant by **genetic engineering**? **[2 marks]**

1 2 . 2 Name the types of enzyme labelled **P** and **Q**. **[2 marks]**

1 2 . 3 Explain why the same enzyme is used to cut the insulin gene and the plasmid. **[2 marks]**

1 2 . 4 Explain why the bacteria multiply in a fermenter. **[1 mark]**

1 2 . 5 Describe **one** benefit and **one** risk of genetically modified crops. **[2 marks]**

1 2 . 6 Scientists have used genetic engineering to treat inherited disorders such as cystic fibrosis. Describe how this works and suggest **one** concern. **[3 marks]**

1 3 Cloning produces genetically identical organisms.

1 3 . 1 Name **two** ways to clone a plant. **[2 marks]**

1 3 . 2 Describe how tissue culture is used to clone a rare orchid. **[4 marks]**

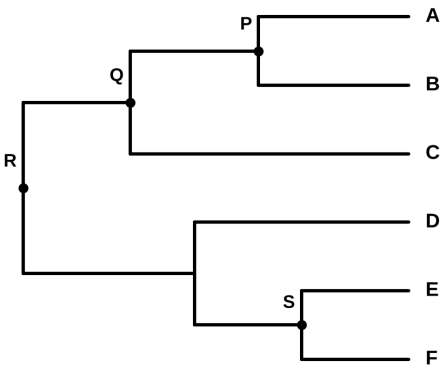
1 3 . 3 Explain **one** benefit and **one** risk of cloning plants for farmers. **[2 marks]**

1 3 . 4 Describe how an embryo transplant can be used to produce many cattle with desirable characteristics. **[3 marks]**

1 3 . 5 Describe the steps of adult cell cloning, as used to produce Dolly the sheep. **[4 marks]**

1 3 . 6 Suggest **two** concerns about cloning animals. **[2 marks]**

1 4 Scientists classify organisms. The diagram is an evolutionary tree for six species, **A** to **F**. The letters **P** to **S** show points where species split.



1 4 . 1 Which two species are the most closely related? Give a reason. **[2 marks]**

1 4 . 2 Which point is the common ancestor of **all six** species? **[1 mark]**

1 4 . 3 Species **C** is more closely related to **A** than to **D**. Explain how the diagram shows this. **[2 marks]**

1 4 . 4 Linnaeus developed a system of naming species using two names, such as *Panthera leo* for a lion. What is the name of this system? **[1 mark]**

1 4 . 5 In *Panthera leo*, state which word is the genus. **[1 mark]**

1 4 . 6 State the three domains used in the three-domain system proposed by Carl Woese. **[3 marks]**

1 4 . 7 Explain why classification has changed over time. **[2 marks]**

1 5 Fossils provide evidence for evolution.

1 5 . 1 Describe **two** ways in which fossils may be formed. **[2 marks]**

1 5 . 2 Explain why there are few fossils of early forms of life. **[2 marks]**

1 5 . 3 Fossils of a horse show that it has changed over millions of years. Explain how this supports the theory of evolution. **[2 marks]**

1 5 . 4 What is meant by **extinction**? **[1 mark]**

1 5 . 5 Give **three** factors that may cause a species to become extinct. **[3 marks]**

1 5 . 6 Antibiotic resistance and fossil evidence are both used as evidence for evolution. Suggest why scientists can observe evolution in bacteria but not in large animals. **[2 marks]**

1 6 Gregor Mendel carried out breeding experiments on pea plants in the 19th century.

1 6 . 1 Give **two** conclusions Mendel made about inheritance. **[2 marks]**

1 6 . 2 Explain why Mendel's work was not accepted until after his death. **[2 marks]**

1 6 . 3 In the 20th century scientists observed chromosomes during cell division. Describe how this supported Mendel's work. **[2 marks]**

16.4 Name **two** scientists who worked out the structure of DNA in 1953. **[2 marks]**

17 Stretch questions (grade 7–9).

17.1 *Explain how natural selection can lead to the formation of a new species, using an example of two populations separated by a geographical barrier. **[6 marks]**

17.2 *Compare selective breeding and genetic engineering. Evaluate their use in producing crops with higher yields. **[6 marks]**

1 7 . 3 *Explain how sexual reproduction, meiosis and mutation lead to variation and how this gives a species the ability to survive when the environment changes. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.6: Inheritance, variation and evolution

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	sexual: two parents / asexual: one parent ; sexual: gametes (sex cells) are involved / asexual: no gametes ; sexual: offspring genetically different ; asexual: offspring genetically identical (clones) ;	One mark for each correct row (both cells needed). Allow any four correct statements.	4	AO1 4.6.1.1
01.2	meiosis		1	AO1 4.6.1.2
01.3	produces variation in the offspring ; if the environment changes, some offspring may survive / natural selection can act ;	Allow: can be used in selective breeding.	2	AO1 4.6.1.1
01.4	only one parent needed / no need to find a mate ; faster / many identical offspring produced ;	Allow: uses less energy; identical offspring are well adapted to a stable environment.	2	AO1 4.6.1.1
01.5	malaria protists / fungi / strawberry plants / daffodils ; binary fission / spores / runners / bulb division ;	The method must match the example.	2	AO1 4.6.1.1
01.6	asexual reproduction produces large numbers of identical offspring quickly ; sexual reproduction increases variation so some survive in a changed environment ;		2	AO3 4.6.1.1
02.1	the DNA is copied / replicated ;		1	AO2 4.6.1.2
02.2	2		1	AO2 4.6.1.2
02.3	23		1	AO1 4.6.1.2
02.4	the DNA is copied then the cell divides twice ; the chromosomes in each pair are separated and shuffled randomly ; producing four gametes with half the number of chromosomes and different combinations of alleles ;		3	AO2 4.6.1.2
02.5	the number of chromosomes is restored to the full number (e.g. 23 + 23 = 46) ; so the new cell has genes from both parents / the correct number of chromosomes ;		2	AO1 4.6.1.2
02.6	meiosis produces gametes with different combinations of chromosomes / alleles ; any sperm can fertilise any egg so many possible combinations ;		2	AO2 4.6.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	mitosis produces two cells, meiosis produces four ; mitosis produces identical cells (diploid), meiosis produces genetically different cells with half the chromosomes (haploid) ;		2	AO1 4.6.1.2
02.8	ovaries and testes (reproductive organs)		1	AO1 4.6.1.2
03.1	double helix		1	AO1 4.6.1.3
03.2	P: (sugar-phosphate) backbone / strand ; Q: base pair (held by hydrogen bonds) ; R: base ;	Allow nucleotide for P only if the answer is clearly the unit of sugar, phosphate and base.	3	AO1 4.6.1.3
03.3	A pairs with T ; C pairs with G ;		2	AO1 4.6.1.3
03.4	nucleotides		1	AO1 4.6.1.3
03.5	the entire genetic material / all the DNA of that organism		1	AO1 4.6.1.3
03.6	search for genes linked to different diseases ; understand and treat inherited disorders ; trace human migration patterns / predict response to drugs ;	Any three.	3	AO3 4.6.1.3
03.7	1200 ÷ 3 ; 400 ;		2	AO2 4.6.1.3
03.8	the order of bases in the gene is a code ; every three bases code for one amino acid ; the amino acids are joined in the order specified to make the protein ;		3	AO1 4.6.1.3
03.9	a different amino acid may be coded ; the protein folds into a different shape so it may not function (e.g. enzyme active site) ; the genetic code is degenerate: the same amino acid may be coded so no change ;	Higher Tier / triple. Allow 'no effect if in non-coding DNA'.	3	AO3 4.6.1.3
04.1	allele ; dominant ; recessive ; genotype ; phenotype ; homozygous ;		6	AO1 4.6.1.4
04.2	heterozygous		1	AO1 4.6.1.4
04.3	nucleus		1	AO1 4.6.1.3
04.4	a recessive allele is expressed when two copies are present (homozygous recessive) ; e.g. cystic fibrosis / blue eyes occur with two recessive alleles ;		2	AO3 4.6.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	B B / B b ; B b / b b ;	Top row: BB, Bb. Bottom row: Bb, bb. One mark for correct gametes (already shown), one mark for all four genotypes correct.	2	AO2 4.6.1.4
05.2	0.25 / 25 % / 1 in 4 / ¼		1	AO2 4.6.1.4
05.3	3 : 1		1	AO2 4.6.1.4
05.4	80 × 0.25 ; 20 ;		2	AO2 MS 1c
05.5	Bb ; the child must inherit one b from each parent / man must carry a b allele ;		2	AO3 4.6.1.4
05.6	TT ; if it were Tt about half the offspring would be dwarf ;		2	AO3 4.6.1.4
05.7	both parents may be carriers / heterozygous (Bb) ; so 25 % chance of a child with bb (blue eyes) ;		2	AO3 4.6.1.4
06.1	ff		1	AO2 4.6.1.6
06.2	P and Q are both carriers / heterozygous (Ff) ; each passes on the recessive allele (f) to S ; S inherits two copies of the recessive allele ;		3	AO3 4.6.1.6
06.3	gametes F and f for each parent ; FF, Ff, Ff, ff ;		2	AO2 4.6.1.4
06.4	¼ / 0.25 / 25 %		1	AO2 4.6.1.4
06.5	R is FF, Ff or Ff (3 possible unaffected outcomes) ; 2/3 ;	Higher Tier.	2	AO3 4.6.1.4
06.6	both parents carriers: 1/25 × 1/25 = 1/625 ; × ¼ ; 1/2500 (0.0004) ;	Higher Tier.	3	AO2 4.6.1.6
06.7	parent may be heterozygous (Pp) ; child can inherit the recessive allele from both parents (pp) ;		2	AO3 4.6.1.6
07.1	K ; has an X and a Y chromosome ;		2	AO2 4.6.1.5
07.2	gametes X, X and X, Y shown ; XX, XX, XY, XY ;		2	AO2 4.6.1.5
07.3	0.5 / ½ / 50 %		1	AO2 4.6.1.5
07.4	no ; each fertilisation is independent / probability is always 0.5 ;		2	AO3 4.6.1.5
07.5	0.5 × 0.5 × 0.5 ; 0.125 / 1/8 / 12.5 % ;		2	AO2 MS 2d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6	all eggs contain an X chromosome ; sperm can carry an X or a Y ;		2	AO2 4.6.1.5
08.1	170–179 cm		1	AO3 WS 3.1
08.2	24 + 31 = 55 ; 55 ÷ 80 × 100 ; 68.75 % (allow 69 %) ;		3	AO2 MS 1c
08.3	a characteristic that can have any value in a range / shows a gradual spread ;		1	AO1 4.6.2.1
08.4	genes inherited from parents ; environment (e.g. diet / nutrition) ;		2	AO1 4.6.2.1
08.5	they have the same genes (genetically identical) ; so the difference is due to the environment (e.g. diet) ;		2	AO3 4.6.2.1
08.6	only genes: blood group / eye colour / inherited disorders ; genes and environment: height / mass / skin colour (tan) ;		2	AO1 4.6.2.1
08.7	random change in the DNA / a gene ; may produce a different protein / new allele ; a very few mutations lead to a new phenotype that is better suited so is passed on (natural selection) ;		3	AO2 4.6.2.1
09.1	increased from 10 % ; to a maximum of about 97 % in 1920 ; fairly constant between 1900 and 1950 ;	Allow data quotes with units.	3	AO3 4.6.2.2
09.2	moths are eaten by birds (predation) ; dark moths were better camouflaged on dark trees so survived ; survivors reproduced and passed on the allele for dark colour ; so the allele frequency of the dark form increased over generations ;		4	AO2 4.6.2.2
09.3	70 − 5 = 65 ; 65 ÷ 70 × 100 ; 92.9 % (93 %) ;		3	AO2 MS 1c
09.4	trees became lighter / lichens returned ; pale moths now better camouflaged so more survive ;		2	AO3 4.6.2.2
09.5	organisms with advantageous features survive and breed ; the environment (selection pressure) changed so the population changed ;		2	AO3 4.6.3.1
09.6	it challenged religious views (God created all species) ; insufficient evidence at the time / the mechanism of inheritance (genes) was unknown ;	Allow: it took a long time to accumulate evidence; Wallace's work.	2	AO1 4.6.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.1	a change in the inherited characteristics of a population ; over time / many generations (through natural selection) ;		2	AO1 4.6.2.2
10.2	populations become isolated (e.g. by a geographical barrier) ; different environments → different selection pressures ; the populations accumulate different alleles / phenotypes ; eventually they can no longer interbreed to produce fertile offspring (new species) ;	Triple only.	4	AO2 4.6.3.1
10.3	random mutation produces a resistant bacterium ; antibiotic kills non-resistant bacteria so the resistant ones survive and reproduce ; resistant strain spreads ;		3	AO2 4.6.3.3
10.4	MRSA is resistant to many antibiotics ;		1	AO2 4.6.3.3
10.5	do not use antibiotics for viral infections / unnecessarily ; complete the full course ;		2	AO1 4.6.3.3
11.1	humans choose organisms with desired characteristics ; and breed them together over many generations ;		2	AO1 4.6.2.3
11.2	select cows with the highest milk yield ; select bulls whose mothers had high yield ; breed them together ; select the offspring with high yield and repeat for many generations ;		4	AO2 4.6.2.3
11.3	36 − 20 = 16 ; 16 ÷ 20 × 100 ; 80 % ;		3	AO2 MS 1c
11.4	16 ÷ 8 ; 2 dm^3 per day per generation ;		2	AO2 MS 1a
11.5	limit to what the breed can produce / genetic variation is reduced ;		1	AO3 4.6.2.3
11.6	reduced gene pool / less variation ; more chance of inherited defects / disease and the population may not cope with changes ;		2	AO2 4.6.2.3
11.7	disease resistance in food crops ; animals that produce more meat / large or unusual flowers / gentle nature of dogs ;	Any two sensible examples.	2	AO1 4.6.2.3
12.1	modifying the genome of an organism ; by introducing a gene from another organism to give a desired characteristic ;		2	AO1 4.6.2.4
12.2	P: restriction enzyme ; Q: ligase ;	Triple only.	2	AO1 4.6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.3	it leaves the same (complementary) sticky ends ; so the gene fits into / joins with the plasmid ;	Higher Tier.	2	AO2 4.6.2.4
12.4	to produce large amounts of insulin ;		1	AO2 4.6.2.4
12.5	benefit: increased yield / pest or disease resistance / improved nutrition (e.g. golden rice) ; risk: effect on wild flowers / insects / human health uncertain ;		2	AO3 4.6.2.4
12.6	gene therapy: working copies of the gene are put into cells of the patient (e.g. lung cells) ; to replace the faulty allele ; concern: effects on offspring / safety / cost / ethical issues ;	Triple only (tests of understanding of ethical issues).	3	AO3 4.6.2.4
13.1	taking cuttings ; tissue culture (micropropagation) ;		2	AO1 4.6.2.5
13.2	take small pieces of tissue (from the shoot tip) ; grow them on sterile nutrient agar containing hormones ; growth into small plantlets ; plantlets then transferred to compost ;		4	AO2 4.6.2.5
13.3	benefit: quick / produces many identical plants with a good characteristic ; risk: less variation so all can be wiped out by one disease ;		2	AO3 4.6.2.5
13.4	sperm from a prize bull and eggs from a prize cow used to produce embryos ; embryo split into several cells before they become specialised ; each embryo placed into a surrogate cow ;		3	AO1 4.6.2.5
13.5	nucleus removed from an egg cell ; nucleus from an adult body cell inserted into the empty egg ; an electric shock stimulates the egg to divide to form an embryo ; embryo is placed in the womb of a surrogate ;	Triple only.	4	AO1 4.6.2.5
13.6	low success rate / suffering of the animals ; reduced variation / ethical concerns about cloning humans ;		2	AO3 4.6.2.5
14.1	A and B ; they share the most recent common ancestor (P) ;		2	AO3 4.6.3.4
14.2	R		1	AO3 4.6.3.4
14.3	C and A share a common ancestor (Q) more recently ; than the ancestor shared with D (R) ;		2	AO3 4.6.3.4
14.4	binomial system		1	AO1 4.6.3.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
14.5	Panthera		1	AO1 4.6.3.4
14.6	Archaea ; Bacteria ; Eukaryota ;	Triple only.	3	AO1 4.6.3.4
14.7	new evidence from chemical analysis / DNA sequencing ; better microscopes show internal structures / evolutionary relationships ;		2	AO2 4.6.3.4
15.1	hard parts (bones, shells) replaced by minerals as they decay ; preserved in ice / amber / peat / tar pits where no decay happens ; (or) casts / impressions such as footprints ;	Any two.	2	AO1 4.6.3.2
15.2	early organisms were soft-bodied so left few traces ; traces destroyed by geological activity (heat, pressure) ;		2	AO2 4.6.3.2
15.3	fossils in older rock layers are more different from modern horses ; shows gradual change as species evolved / adapted ;		2	AO3 4.6.3.2
15.4	no individuals of a species remain alive ;		1	AO1 4.6.3.2
15.5	new predators / new diseases / new competitors ; environmental changes (e.g. climate change, habitat loss) ; catastrophic events (e.g. volcanic eruptions, asteroid impact) ;		3	AO1 4.6.3.2
15.6	bacteria reproduce very quickly (short generation time) ; so many generations / changes can be seen in a short time ;		2	AO3 4.6.3.3
16.1	characteristics are determined by units (now called genes) ; units can be dominant or recessive / each parent passes one unit to offspring ;		2	AO1 4.6.3.5
16.2	scientists at the time had not observed chromosomes / genes ; his work was not understood / no mechanism (cell division) known ;		2	AO3 4.6.3.5
16.3	chromosomes behave in the same way as the 'units' (separate and pair up) ; so genes were found to be on chromosomes ;		2	AO1 4.6.3.5
16.4	Watson ; Crick ;	Allow Franklin / Wilkins for data.	2	AO1 4.6.3.5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
17.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • variation exists in a population due to mutation / sexual reproduction • two populations separated by a barrier (e.g. sea / mountains) cannot interbreed • each experiences different environmental conditions (selection pressures) • individuals with advantageous alleles survive and breed; those alleles become more common • over many generations the populations differ greatly in their genotypes / phenotypes • if they can no longer interbreed to produce fertile offspring they are different species (speciation)		6	AO2 4.6.3.1
17.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • selective breeding: choose parents with the desired features; breed over many generations; slow • genetic engineering: transfer a gene from another organism into the crop cells; quick and precise • selective breeding reduces variation (gene pool); risk of inbreeding and loss of resistance • genetic engineering can add genes from different species (e.g. pest resistance, golden rice) • concerns about GM: effects on food chains / wild plants, possible effects on health, cost, patents • balanced judgement, e.g. both can raise yield; GM faster but needs regulation; selective breeding is safer but slower		6	AO3 4.6.2.3 / 4.6.2.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
17.3	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • mutations are random changes in DNA that create new alleles • meiosis shuffles chromosomes producing gametes with different combinations of alleles • random fertilisation: any sperm can fertilise any egg giving many different genotypes • offspring differ from parents and from each other • when the environment changes, some individuals have features that give an advantage • they survive and reproduce so the species continues (natural selection); asexual clones have less chance		6	AO2 4.6.1.2 / 4.6.2.1

Where marks are lost in Topic 4.6

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Meiosis** makes four genetically different gametes with half the chromosomes (23); **mitosis** makes two identical cells with 46. Learn the numbers.
- ☐ **Key terms:** allele, dominant, recessive, homozygous, heterozygous, genotype, phenotype. Use them in every genetics answer.
- ☐ **Punnett squares:** write the parent genotypes, write the gametes, fill the grid, then **read off the ratio / probability**. Heterozygous × heterozygous = 3 : 1 (25 % recessive).
- ☐ **Pedigree trick:** two unaffected parents with an affected child → the allele is recessive and both parents are carriers.
- ☐ **Probability:** independent events multiply (0.5 × 0.5 × 0.5). The sex of the next child is always 50 : 50.
- ☐ **Natural selection:** variation → better-adapted survive → reproduce → pass on alleles → allele becomes more frequent. Always include all four ideas.
- ☐ **Selective breeding vs genetic engineering:** breeding = many generations, choose parents; genetic engineering = transfer a gene (restriction enzyme cuts, ligase joins).
- ☐ **Evolutionary trees:** the more recent the shared ancestor, the closer the relationship. Read the branch points, not the order of the species at the tips.
- ☐ **Classification:** binomial name = genus + species (Genus capital, species lower-case). Three domains: Archaea, Bacteria, Eukaryota.
- ☐ **Percentage change = change ÷ original × 100.** For '% increase' always divide by the starting value.